

STAT5147 Module 3 Homework

Aheer Sravon

Used Function(s)

```
require(ggplot2)
```

Loading required package: ggplot2

```
plot.discrete <- function(y, proby) {  
  bars <- as.data.frame(cbind(y, proby))  
  
  ggplot(bars, aes(x=y, y=proby)) +  
    geom_bar(stat="identity", width=1, fill="blue", colour="black") +  
    theme_bw() +  
    theme(axis.title.y=element_text(size=rel(1.4)),  
          axis.title.x=element_text(size=rel(1.4)),  
          axis.text.x=element_text(size=rel(1.2)),  
          axis.text.y=element_text(size=rel(1.2)),  
          plot.title=element_text(hjust=0.5, size=rel(1.6))) +  
    labs(  
      x="y",  
      y="p(y)",  
      title="Distribution of Discrete R.V."  
    )  
}
```

```
plot.binom <- function(n, p) {  
  y <- c(0:n)  
  proby <- dbinom(0:n, n, p)  
  bars <- as.data.frame(cbind(y, proby))  
}
```

```

ggplot(bars, aes(x=y, y=proby)) +
  geom_bar(stat="identity", width=1, fill="blue", colour="black") +
  theme_bw() +
  theme(axis.title.y=element_text(size=rel(1.4)),
        axis.title.x=element_text(size=rel(1.4)),
        axis.text.x=element_text(size=rel(1.2)),
        axis.text.y=element_text(size=rel(1.2)),
        plot.title=element_text(hjust=0.5, size=rel(1.6))) +
  labs(x="Number of Successes (y)",
       y="Probability",
       title="Distribution of Binomial R.V.")
}

plot.pois<- function(lambda){
  maxy<- qpois(0.9999, lambda)
  y<- c(0:maxy)
  proby<- dpois(0:maxy, lambda)
  Bars<- as.data.frame(cbind(y, proby))
  ggplot(Bars, aes(x = y, y = proby))+
    geom_bar(stat="identity", width = 1,
            fill = "blue", colour = "black")+
    theme_bw()+
    theme(axis.title.y = element_text(size = rel(1.4)),
          axis.title.x = element_text(size = rel(1.4)),
          axis.text.x = element_text(size = rel(1.2)),
          axis.text.y = element_text(size = rel(1.2)),
          plot.title = element_text(hjust=0.5, size = rel(1.6)))+
    labs(x = "Number of Events (y)",
         y = "Probability",
         title = "Distribution of Poisson R.V.")
}

```

Problem 5

Answer to the question e is calculated below,

```
sum(dbinom(60:100, 100, 1/2))
```

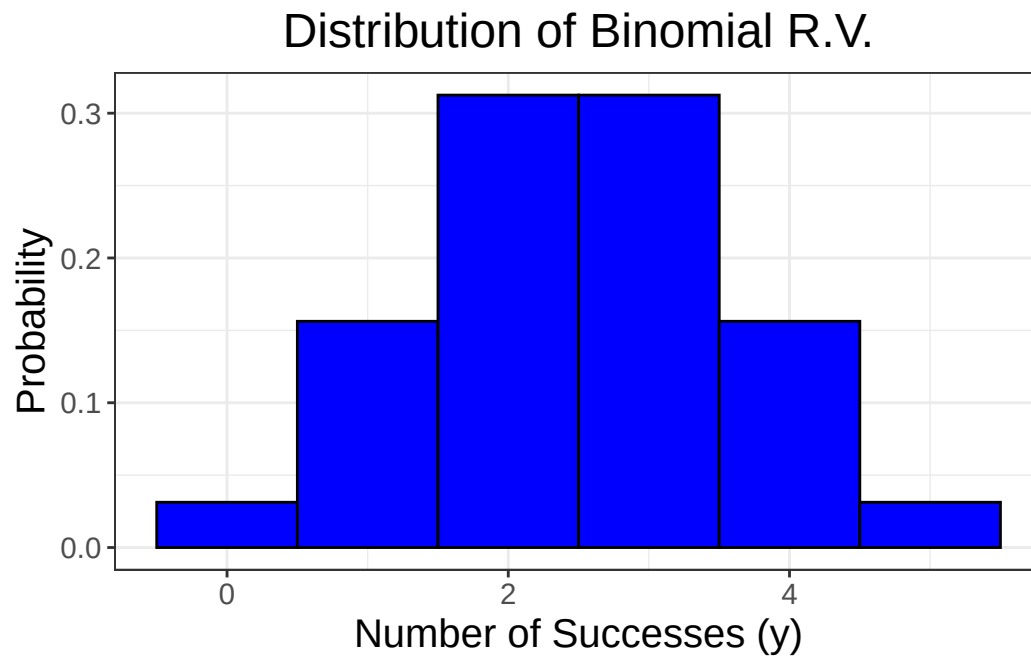
```
[1] 0.02844397
```

Which is only about 3%

Problem 7

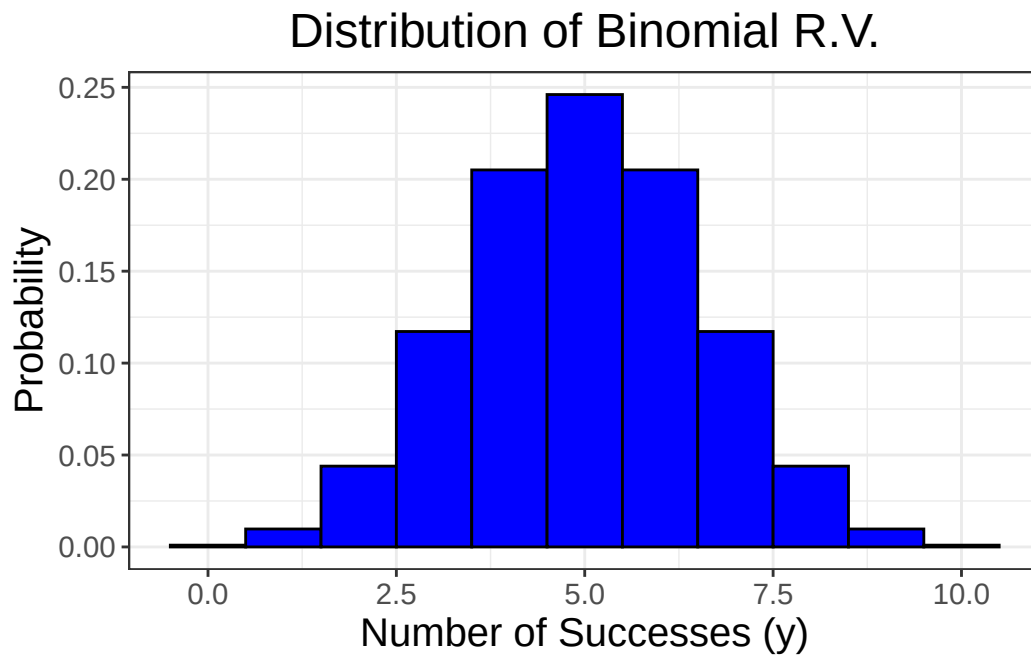
- $p = 0.5, n = 5$

```
plot.binom(5, 0.5)
```



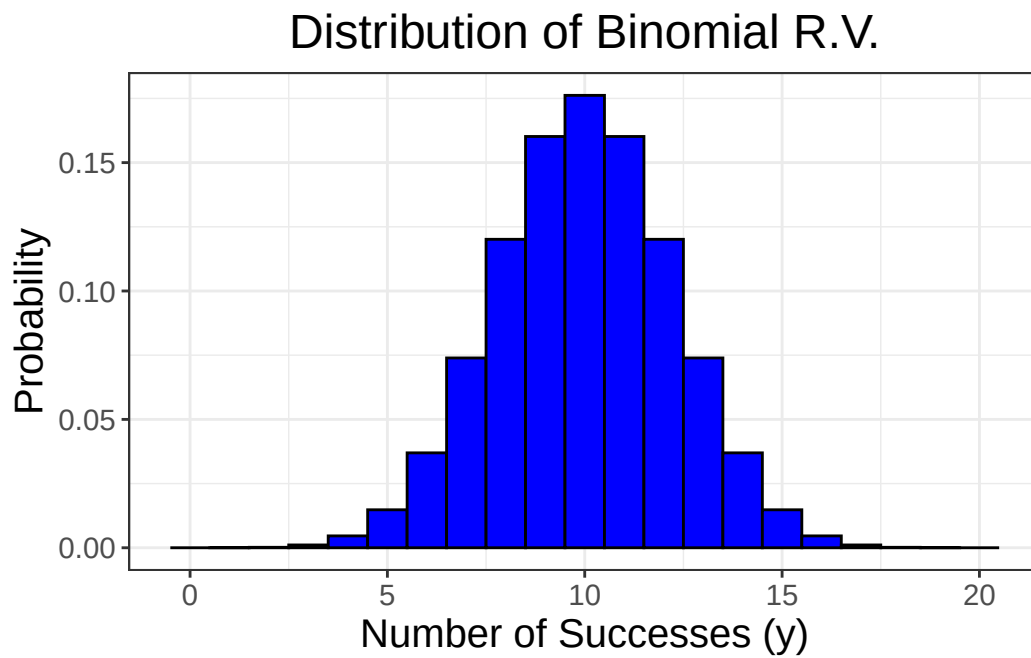
- $p = 0.5, n = 10$

```
plot.binom(10, 0.5)
```



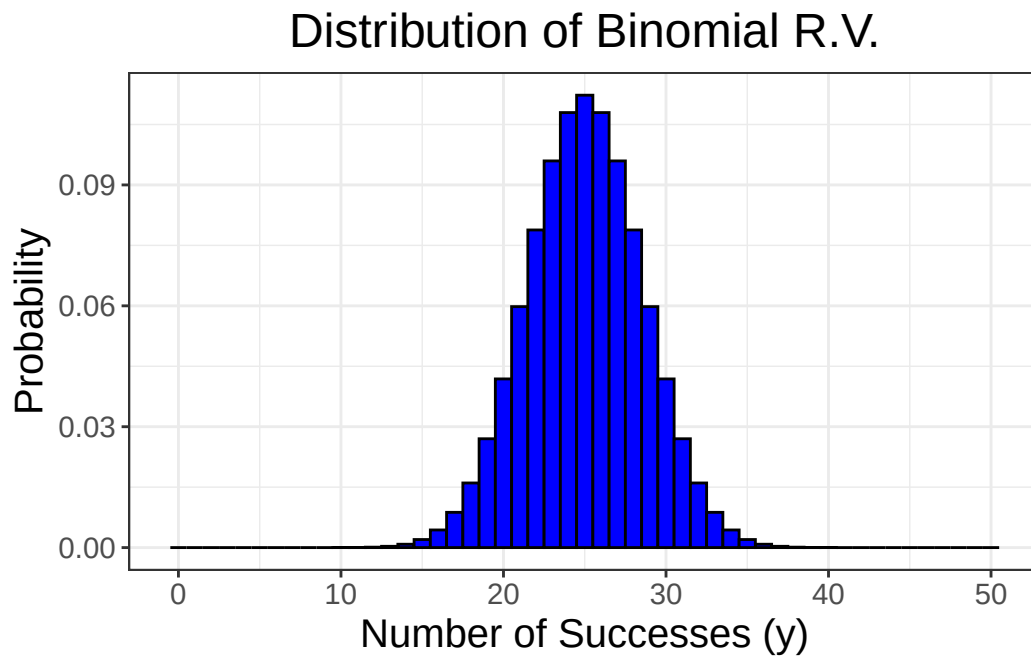
- $p = 0.5, n = 20$

```
plot.binom(20, 0.5)
```



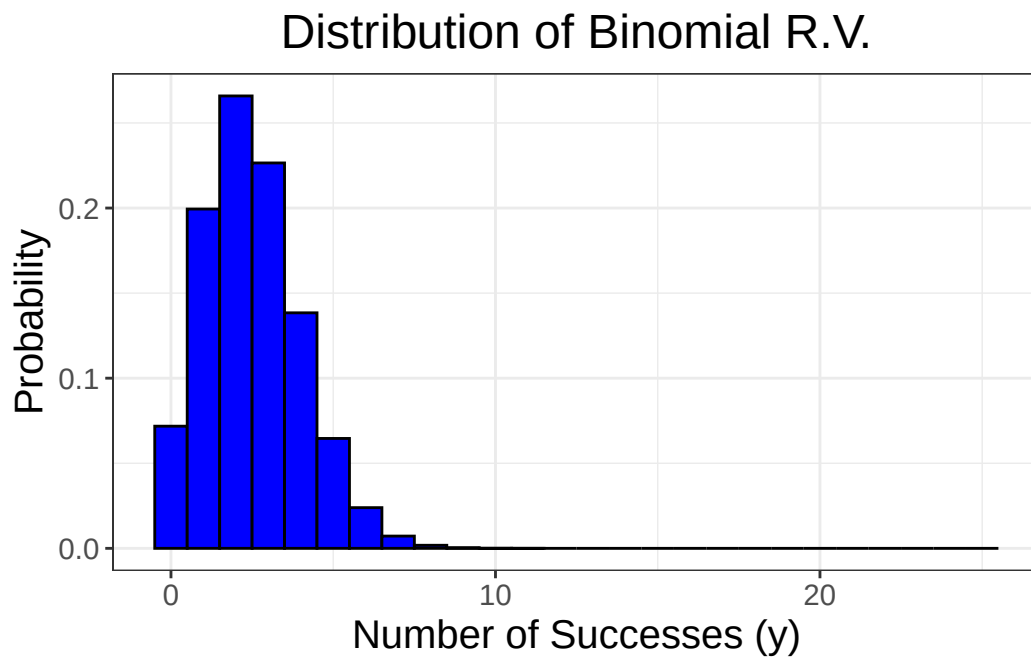
- $p = 0.5, n = 50$

```
plot.binom(50, 0.5)
```



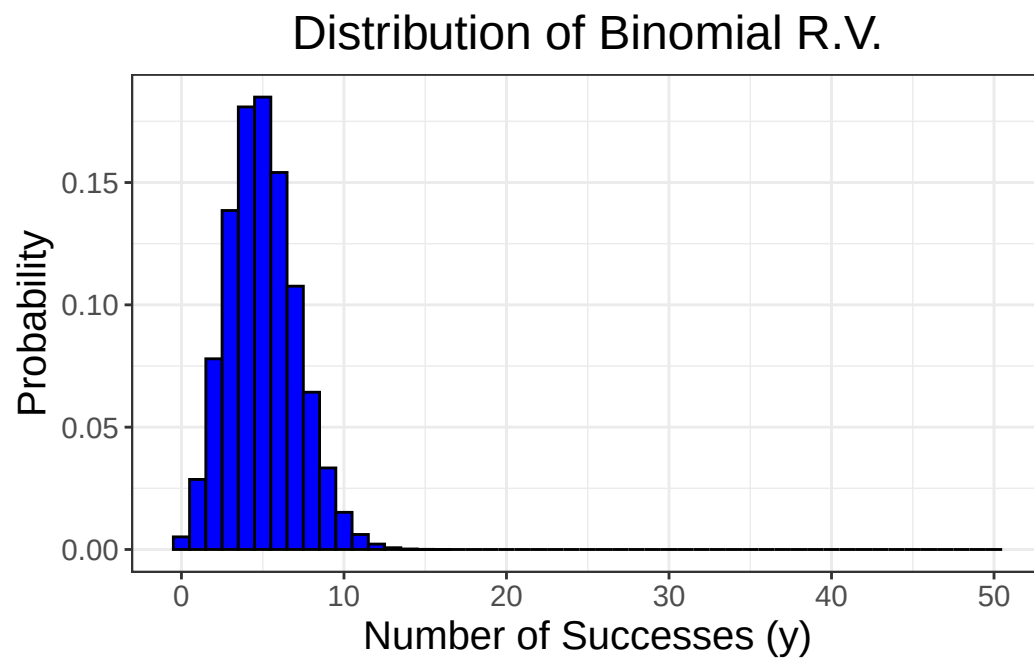
- $p = 0.1, n = 25$

```
plot.binom(25, 0.1)
```



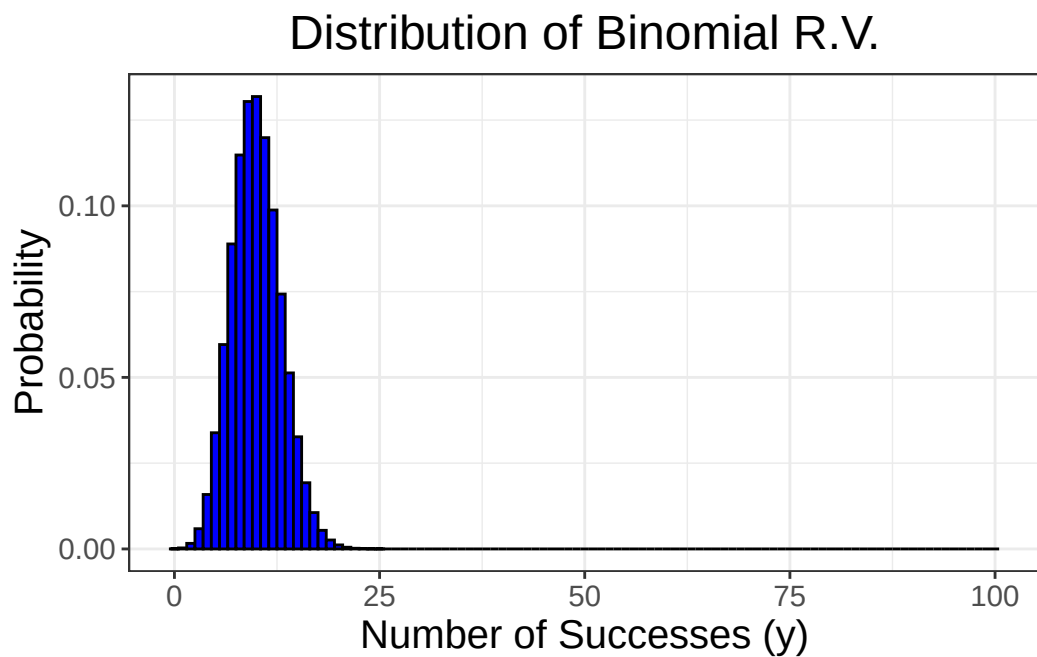
- $p = 0.1, n = 50$

```
plot.binom(50, 0.1)
```



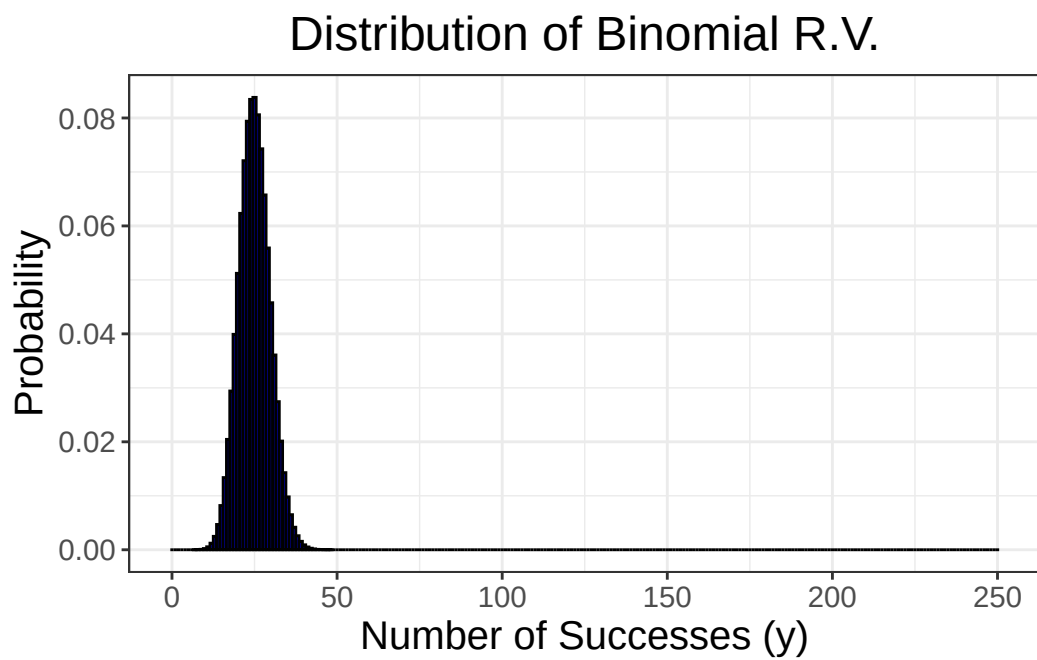
- $p = 0.1, n = 100$

```
plot.binom(100, 0.1)
```



- $p = 0.1, n = 250$

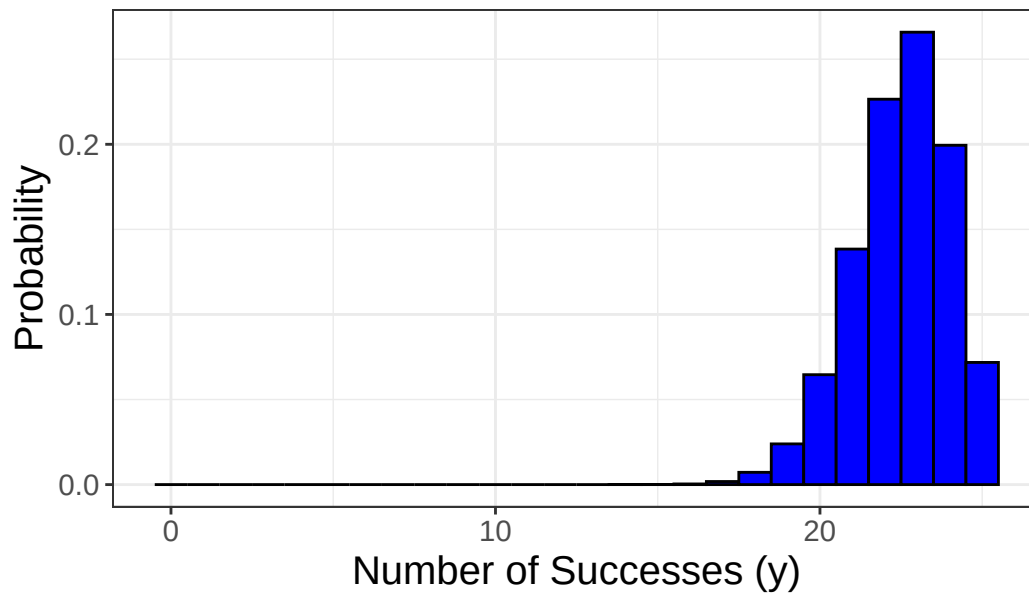
```
plot.binom(250, 0.1)
```



- $p = 0.9, n = 25$

```
plot.binom(25, 0.9)
```

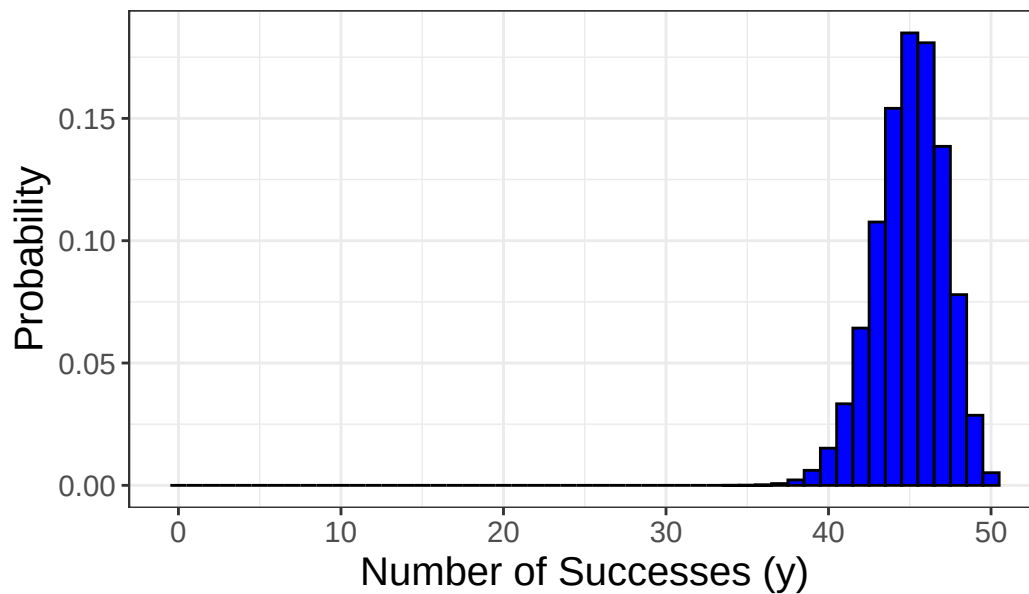
Distribution of Binomial R.V.



- $p = 0.9, n = 50$

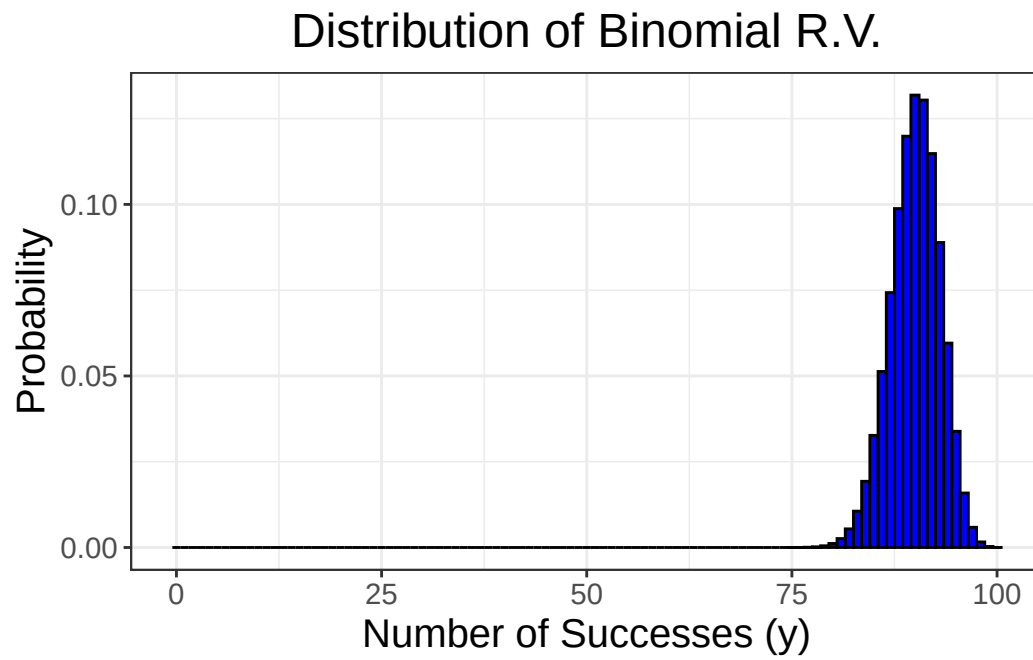
```
plot.binom(50, 0.9)
```

Distribution of Binomial R.V.



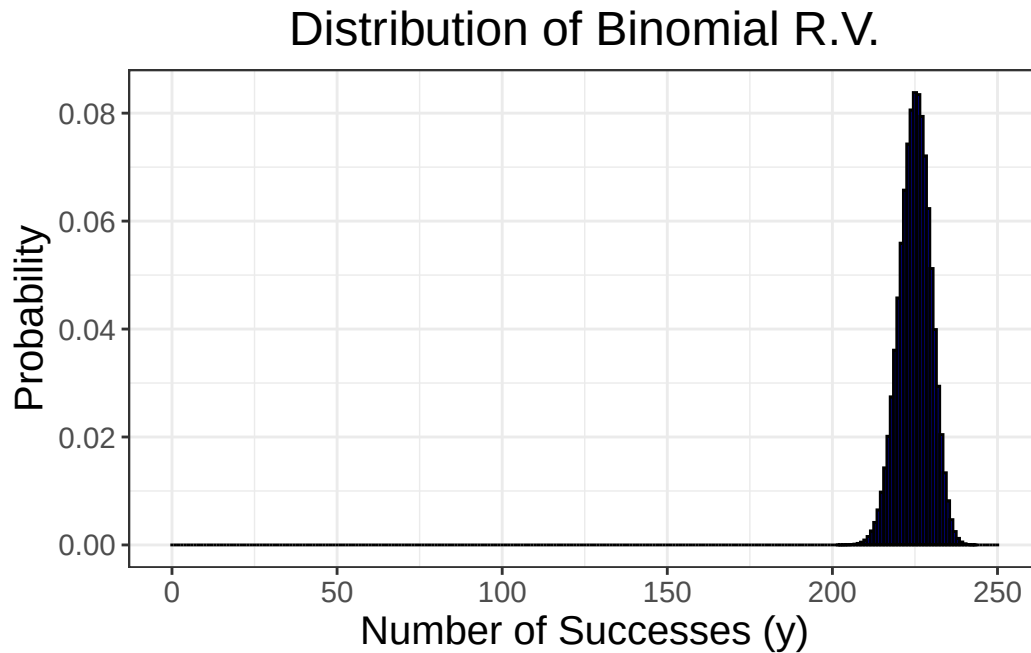
- $p = 0.9, n = 100$

```
plot.binom(100, 0.9)
```



- $p = 0.9, n = 250$

```
plot.binom(250, 0.9)
```



So,

- a) $p = 0.5$
- b) $p = 0.1$ and $p = 0.9$ are mirror images of each other for the same sample size n .
- c) $p = 0.1$
- d) $p = 0.9$
- e) For the last two distributions, the graphs are approximately symmetric for each value of p . As the smaller of np and $n(1 - p)$ increases, the binomial distribution becomes less skewed and more symmetric about its mean.

Problem 8

Poisson distribution with $\lambda = 5$ per hour.

- a) Probability that exactly four arrivals occur during a particular hour is,

```
dpois(4, 5)
```

```
[1] 0.1754674
```

- b) Probability that at most 2 people arrive during a particular hour is,

```
sum(dpois(0:2, 5))
```

```
[1] 0.124652
```

c) Probability that at least 4 people arrive during a particular hour,

```
1 - sum(dpois(0:3, 5))
```

```
[1] 0.7349741
```

d) Both the mean and the variance of the number of people I expect to arrive during a particular hour is 5.

e) It's a binomial distribution with success probability equal to $1 - \text{sum}(\text{dpois}(0:6, 5))$. So the probability is,

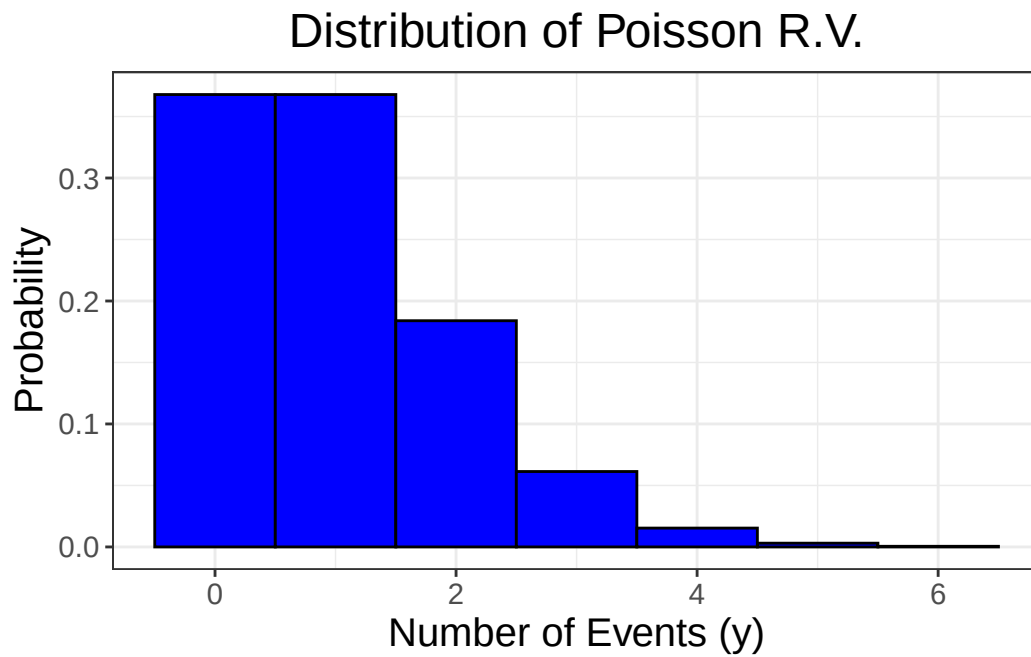
```
(1 - dbinom(0, 8, (1-sum(dpois(0:6, 5)))))
```

```
[1] 0.8861125
```

Problem 9

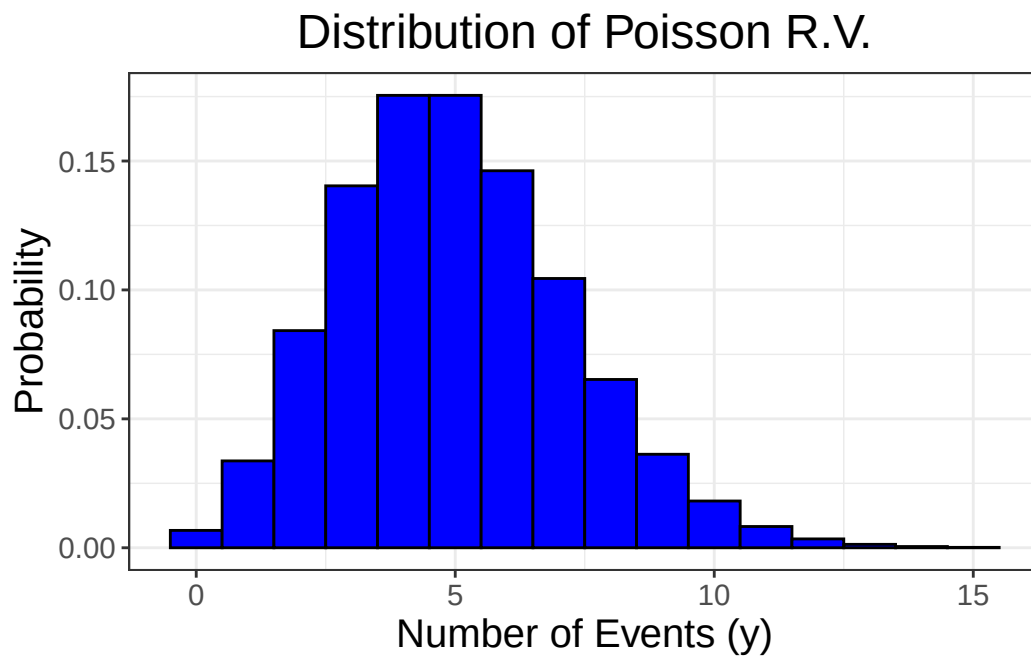
- $\lambda = 1$

```
plot.pois(1)
```



- $\lambda = 5$

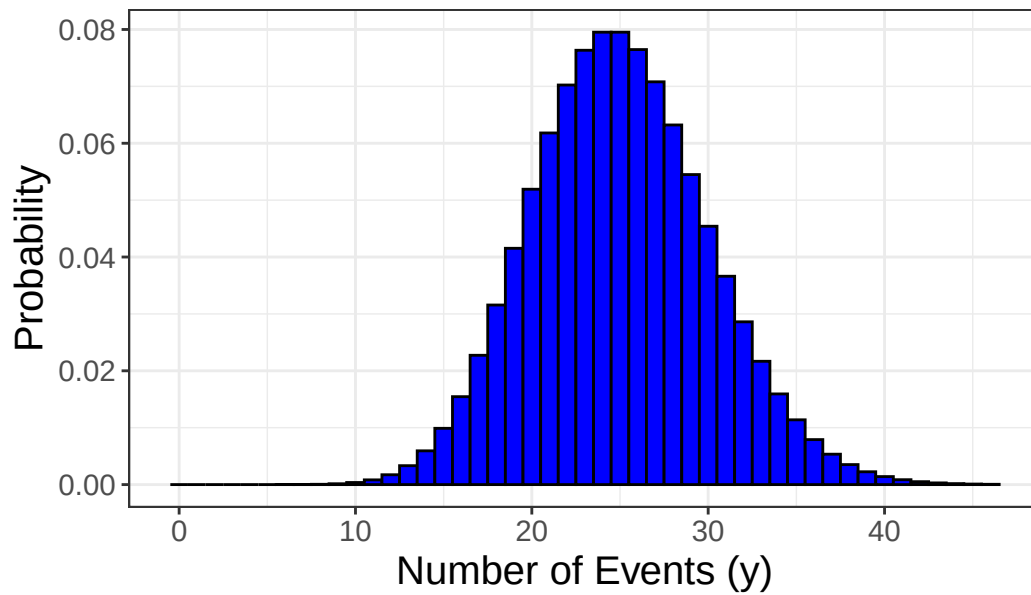
```
plot.pois(5)
```



- $\lambda = 25$

```
plot.pois(25)
```

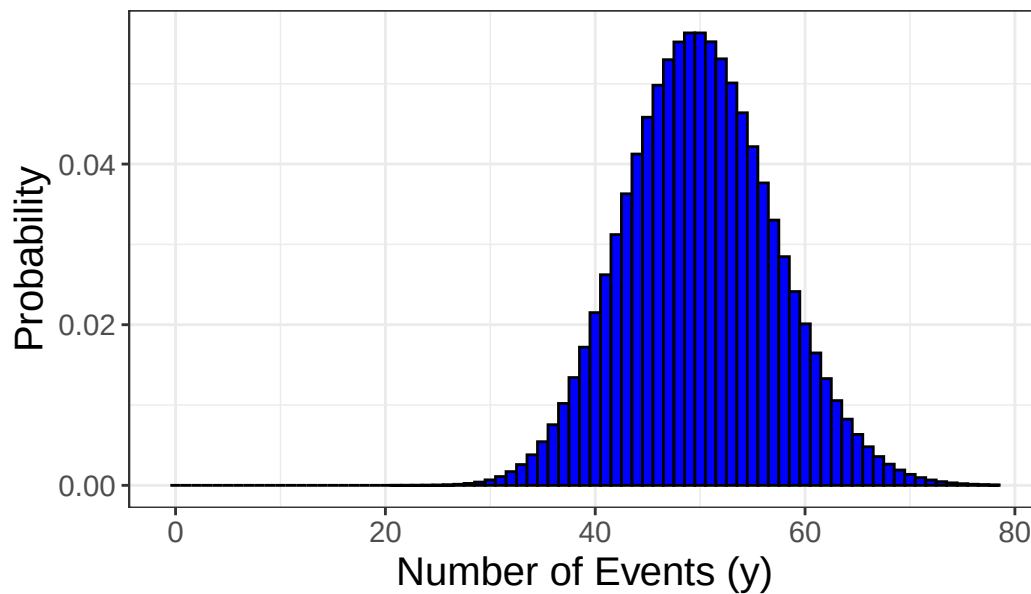
Distribution of Poisson R.V.



- $\lambda = 50$

```
plot.pois(50)
```

Distribution of Poisson R.V.



As λ increases, the Poisson distribution shifts to the right, becomes more spread out, and changes from right-skewed to increasingly symmetric around its mean λ .